

Education

Cornell University

Doctor of Philosophy in Robotics (Mechanical)

Ithaca, NY

Expected Graduation: May 2030

William Marsh Rice University

Bachelor of Science in Mechanical Engineering

Minor in Engineering Design

Houston, TX

Graduation: May 2025

Distinction in Research and Creative Works

Research Interests

- | | | |
|----------------|----------------------|----------------|
| • Robotics | • Mechatronics | • Manipulation |
| • Biomechanics | • Engineering Design | • Augmentation |

Publications

2025

C. Darkes-Burkey, Y. Park, G. Eken, **H. Webb**, R. Shepherd, Controlling Convection in Volumetric Additive Manufacturing for Large Volume Structures at Extreme Throughput, *Advanced Functional Materials*

H. Webb, P. Changrunmaneeikul., S. Yuan, and K. Hang; Wearable Roller Rings to Augment In-Hand Manipulation through Active Surfaces, *IEEE International Conference on Intelligent Robots and Systems (IROS) 2025*

Patents

2024

H. Webb, S. Yuan, and K. Hang; “Wearable Roller Rings to Augment In-Hand Manipulation through Actives Surfaces” - US Provisional Patent: Date of Application (July 3, 2024) [*Non-Provisional Patent Pending*]

Presentations

2025

H. Webb, P. Changrunmaneeikul., S. Yuan, and K. Hang; Wearable Roller Rings to Augment In-Hand Manipulation through Active Surfaces. *IEEE International Conference on Intelligent Robots and Systems (IROS) 2025* [Talk/Poster]

K. Noonan, B. Ramsey, J. Villarreal, **H. Webb**; Automated Screen Printer for Fabrication of Electrodes for Biosensors. Rice University Huff Engineering Design Showcase [Poster]

2024

H. Webb, P. Changrunmaneeikul., S. Yuan, and K. Hang; Wearable Roller Rings to Enable Robot Dexterous In-Hand Manipulation through Active Surfaces. ASME Mechanism and Robotics Committee [Talk/Poster]

H. Webb, P. Changrunmaneeikul., S. Yuan, and K. Hang; Wearable Roller Rings to Enable Robot Dexterous In-Hand Manipulation through Active Surfaces. Texas Regional Robotics Symposium (TEROS) [Poster]

2022

J. Chang, R. Lee, **H. Webb**; Body Cooling Device for the Cooling of Humans in Heat Waves. Rice University Engineering Design Showcase [Poster]

Honors & Awards

Sigma Xi Membership Nomination <i>Sigma Xi Scientific Research Honor Society</i>	<i>Oct. 2025</i>
Graduate School Dean's Scholar Engagement Seminar <i>Cornell University</i>	<i>Sep. 2025</i>
IROS-SDC (Student and Developing Countries) Travel Award <i>IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)</i>	<i>Aug. 2025</i>
Distinction in Research and Creative Works <i>William Marsh Rice University</i>	<i>May 2025</i>
Undergraduate Teaching Fellow, Mechanical Engineering <i>William Marsh Rice University</i>	<i>Jan. 2025</i>
3rd Place, Student Mechanism and Robot Design Competition (SMRDC) <i>American Society of Mechanical Engineers (ASME)</i>	<i>Aug. 2024</i>
ASME Student Financial Support Award <i>American Society of Mechanical Engineers (ASME)</i>	<i>Jun. 2024</i>
Undergraduate Teaching Fellow, Mechanical Engineering <i>William Marsh Rice University</i>	<i>Jan. 2024</i>

Technical Skills

Software: SolidWorks, Onshape, Eagle

Programming Languages: Python, ROS, C++, Matlab

Technical: Product Manufacturing (3D Printing, CNC, Lathe), Embedded System Design, Welding (MIG, Stick)

Tools: Microsoft Office (Word, Excel, Powerpoint), LaTeX

Selected Projects

Automated Screen Printer for Fabrication of Electrodes for Biosensors *Aug. 2024 - May 2025*
Senior Capstone Design Project

- Fully developed an automated screen printer for electrode fabrication in the Sempionatto Research Group capable of taking in an electrode vector file and outputting fully screened sensors with a 95% yield
- Independently designed and constructed a working gantry for movement in all cartesian directions
 - Device's machined parts were fabricated through the use of various subtractive manufacturing methods such as plasma-cutting, waterjet-cutting, and laser-cutting
- Designed a primitive weighted motion planner for ink and screening toolpaths, generating task-space coordinates with motor control flags in G-code format for electronics bus
- Created closed-loop velocity and position control for use in gcode execution and device homing/jogging
- Developed a custom UI for preprocessing electrode vector files, modeled after common 3D printing slicers to streamline tool path generation, repeated electrode Cricut file, and parameter tuning

In-Hand Manipulator for Parallel Jaw Grippers

Aug. 2024 - May 2025

Independent Undergraduate Research

- Designed a robot hand set for parallel jaw grippers to provide full spatial manipulation through the use of active surfaces, deformable fin-rays, and a compliant double blade rotary beam system
- Developed a modular, parameterized backing for quick installation on robots of all levels of fidelity in order to upgrade their manipulation capabilities

Electronic Prony Brake Torquemeter

Jun. 2024 - Aug. 2024

Rice Mechanical Engineering Department

- Designed and developed multiple functional 3D-Printed Prony Brakes to mechanically test the concepts of dynamic torque, speed, and power through inset motors and load cells.
- Built an electronic subsystem consisting of a photo interrupter, Arduino microcontroller, and signal amplifier to calculate the power output of a motor for use in an upper level mechanical engineering course.

Wearable Roller Rings to Augment In-Hand Manipulation through Active Surfaces

Aug. 2023 - May 2024

- Designed in-hand manipulator for both human & robotic grasping systems for differential, non-holonomic, and full spatial manipulation through angled active surfaces
- Created compliant attachment using inverted quatrefoil geometry for zero-cost installation on graspers
- Developed a generalized active surface-based motion model for arbitrary geometry objects

Technical Experience

Lead Engineer, Team Pressing Matters

Aug. 2024 - May 2025

Senior Capstone Design Project

- Oversaw team deadlines and communication with sponsors and collaborators, ensuring proper communication on work between the client and development team throughout the project
- Performed several technical design reviews on mechanical subsections to ensure feasibility and design optimization in terms of modularity, cost, and ease of construction

Co-President

May 2023 - May 2024

Rice Robotics Club

- Managed and aided 10+ subteams and 50+ students on the development of various robotics projects such as several featherweight combat robots and a biomimetic four-legged walking robot
- Developed new safety and testing procedures for R&D and Combat Robotics subteams
- Organized and instructed several workshops, talks, and presentations on robotics and engineering skills

Team Lead, Mirrored Robot Arm Subteam

May 2022 - May 2023

Rice Robotics Club

- Led the development and construction of 6-DoF underactuated robot arm and 17 DoF robot hand
- Promoted collaboration on the integration of manufactured components with electronics subteams to allow for primitive mirroring of human movement in a small-form device

Team Lead, Highschool Aerospace Scholar

Oct. 2019 - Jul. 2020

National Aeronautics and Space Administration (NASA)

- Led 20+ students on the design of a theoretical human landing system between Gateway Space Station and the lunar surface utilizing integrated CAD environment
- Collaborated with other teams to integrate Lunar Lander into theoretically designed Gateway Space Station for mock Artemis Mission

Teaching Experience

Head Teaching Assistant (ENGI 210 - Prototyping & Fabrication)

Jan. 2023 - May 2025

Rice Makerspace, Oshman Engineering Design Kitchen (OEDK)

- Taught material, hosted tutorials, and led workshops on engineering tools and fabrication methods
- Assisted 50+ graduate and undergraduate students with team projects and individual competencies in engineering design

Teaching Assistant (MECH 310 - Rigid Body Dynamics)

Jan. 2025 - May 2025

Rice Undergraduate Teaching Fellow Program, Mechanical Engineering Department

- Assisted with teaching and grading of material for 50+ undergraduate students in an upper level course

Teaching Assistant (MECH 310 - Rigid Body Dynamics)

Jan. 2024 - May 2024

Rice Undergraduate Teaching Fellow Program, Mechanical Engineering Department

- Assisted with teaching and grading of material for 50+ undergraduate students in an upper level course

IBB Summer Academy Assistant

Jun. 2022 - Jul. 2022

Rice Department of Innovation

- Led week-long courses for high school students on engineering design tools, devices, and processes

Mentorship

AP Research Expert Advisor

Aug. 2025 - Present

Conroe High School, Academy for Science and Health Professions (ASHP)

- Lead discussion and assisted high school students with designing, controlling, and building robots for their regional FRC competition (Texas District)

Robotics Club Mentor

May. 2025 - Present

Conroe High School, Academy for Science and Health Professions (ASHP)

- Lead discussion and assisted high school students with designing, controlling, and building robots for their regional FRC competition (Texas District)

FIRST Robotics Competition (FRC) Volunteer

Aug. 2024- May 2025

FRC Texas District

- Aided local competitions in coordinating team movements as well as judging and running game rounds

Relevant Coursework

Algorithmic Robotics, Engineering Design Tools, Foundation of Robotics, Fundamentals of Robotic Manipulation, Fundamentals of Control Systems, Introduction to Operations Research and Optimization, Heat Transfer, Machines and Mechanisms, Modeling Dynamic Systems, Prototyping & Fabrication, Rigid Body Dynamics, Thermodynamics